

- 8 The three distinct roots of the equation  $z^3 - 4z^2 + pz + q = 0$ , where  $p$  and  $q$  are real, are drawn on an Argand diagram. The three points which represent these roots do not lie on a straight line but instead form a triangle  $T$ .

(a) Show that  $T$  is isosceles. [3]

(b) In this question you must show detailed reasoning.

You are given the following information.

- The area of  $T$  is 10 square units.
- One of the roots of the equation  $z^3 - 4z^2 + pz + q = 0$  is  $z = -2$ .

Find the other roots of the equation. [5]